# SOFTWARE ENGINEERING

## Lab7:

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|---|---|---|

## Buggy program:

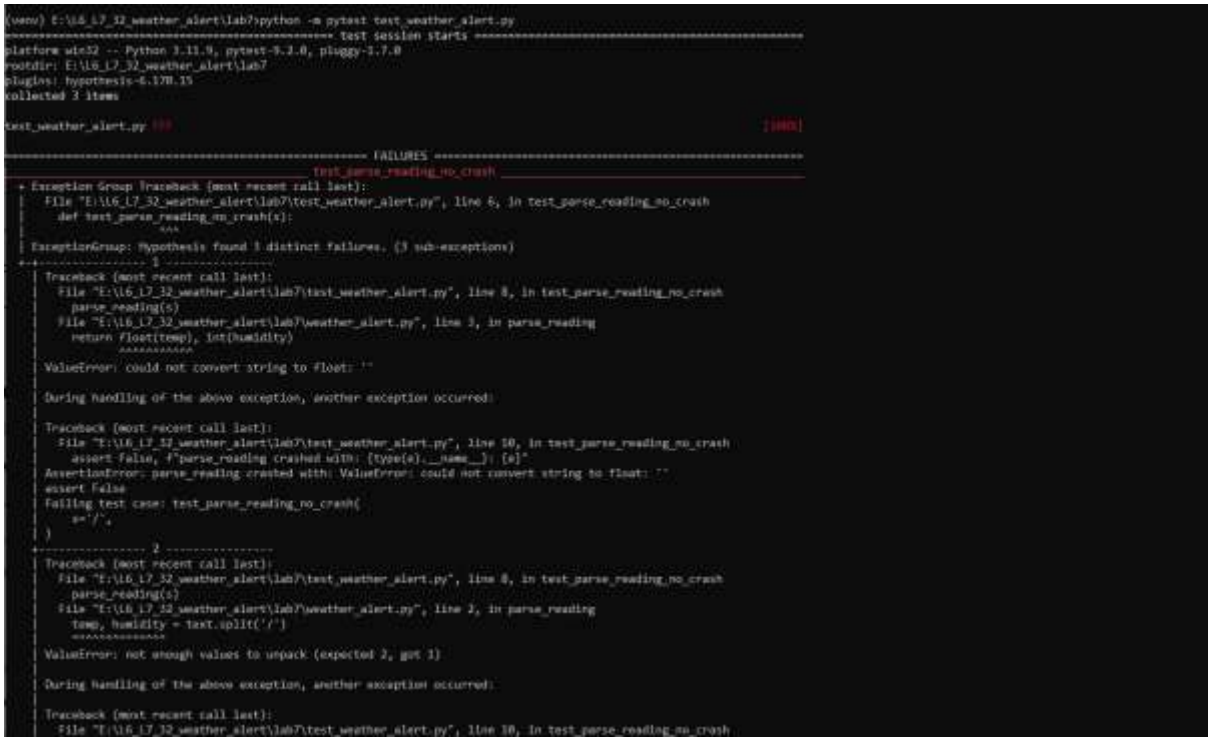
```
(venv) E:\L6_L7_32_weather_alert\lab7>python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): 35.5/80
Temperature and Humidity: (35.5, 80)

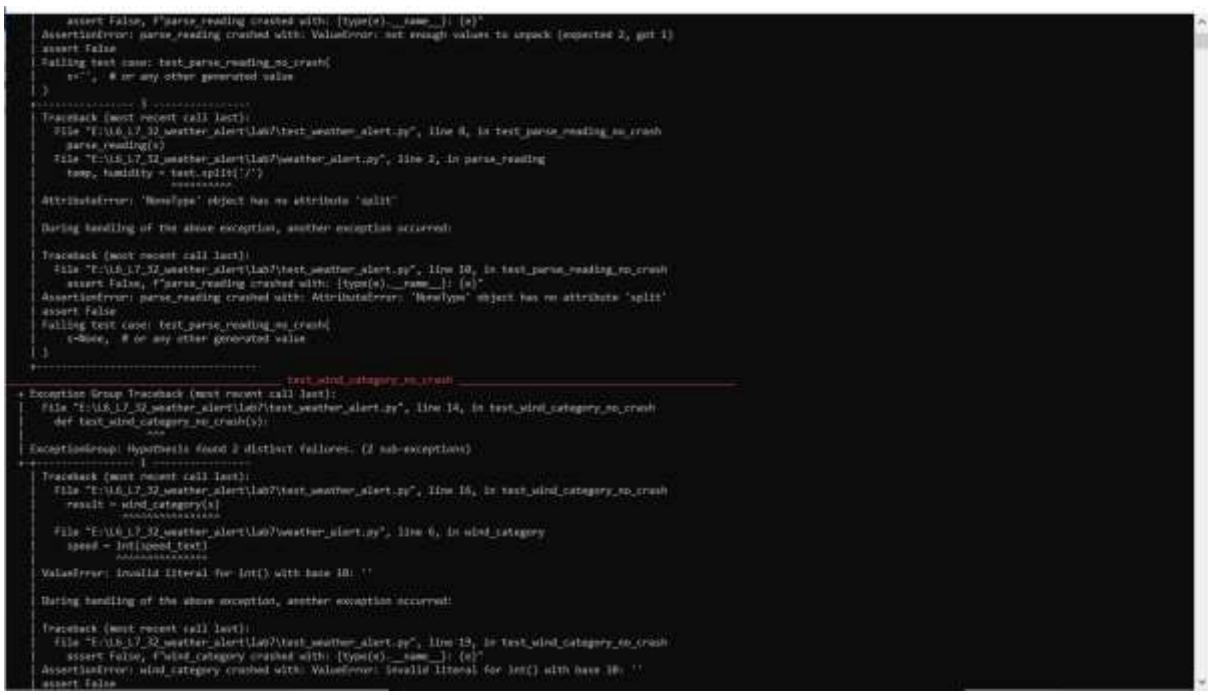
==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): 45
Wind Category: advisory

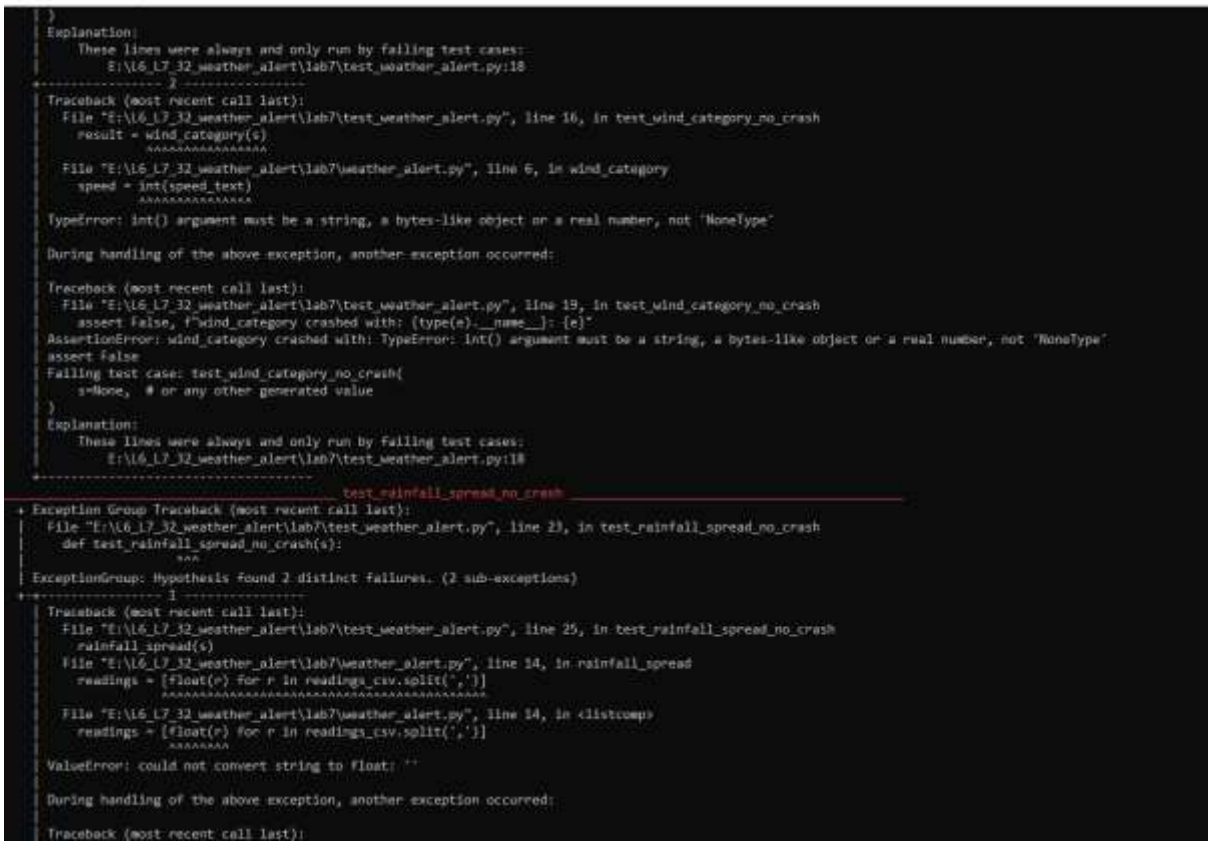
==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 2,5.5,10
Rainfall Spread: 8.0
```

```
(venv) E:\L6_L7_32_weather_alert\lab7>python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): abc
Traceback (most recent call last):
  File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 34, in <module>
    main()
  File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 20, in main
    result = parse_reading(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 2, in parse_reading
    temp, humidity = text.split('/')
    ^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

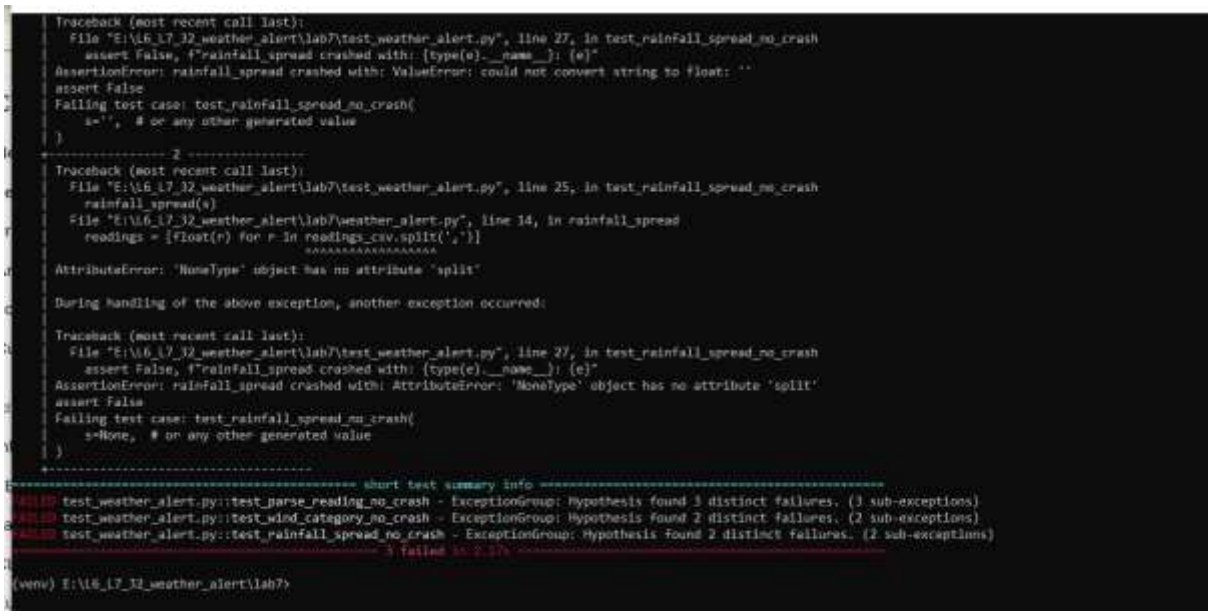
## run the failing tests:





```
    )
    Explanation:
        These lines were always and only run by failing test cases:
            E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py:18
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 16, in test_wind_category_no_crash
    |     result = wind_category(s)
    |              ^^^^^^^^^^^^^^^^
    |   File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 6, in wind_category
    |     speed = int(speed_text)
    |             ^^^^^^^^^^^^^^^
    | TypeError: int() argument must be a string, a bytes-like object or a real number, not 'NoneType'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 19, in test_wind_category_no_crash
    |     assert False, f"wind_category crashed with: {type(e).__name__}: {e}"
    | AssertionError: wind_category crashed with: TypeError: int() argument must be a string, a bytes-like object or a real number, not 'NoneType'
    | assert False
    | Falling test case: test_wind_category_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py:18
    +-----------------------------------
_____ test_rainfall_spread_no_crash _____
+ Exception Group Traceback (most recent call last):
|   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 23, in test_rainfall_spread_no_crash
|     def test_rainfall_spread_no_crash(s):
|                ^^^
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 25, in test_rainfall_spread_no_crash
  |     rainfall_spread(s)
  |   File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 14, in rainfall_spread
  |     readings = [float(r) for r in readings_csv.split(',')]
  |                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 14, in <listcomp>
  |     readings = [float(r) for r in readings_csv.split(',')]
  |                 ^^^^^^^^
  | ValueError: could not convert string to float: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
```



```
  | Traceback (most recent call last):
  |   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 27, in test_rainfall_spread_no_crash
  |     assert False, f"rainfall_spread crashed with: {type(e).__name__}: {e}"
  | AssertionError: rainfall_spread crashed with: ValueError: could not convert string to float: ''
  | assert False
  | Falling test case: test_rainfall_spread_no_crash(
  |     s='',  # or any other generated value
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 25, in test_rainfall_spread_no_crash
  |     rainfall_spread(s)
  |   File "E:\L6_L7_32_weather_alert\lab7\weather_alert.py", line 14, in rainfall_spread
  |     readings = [float(r) for r in readings_csv.split(',')]
  |                                   ^^^^^^^^^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "E:\L6_L7_32_weather_alert\lab7\test_weather_alert.py", line 27, in test_rainfall_spread_no_crash
  |     assert False, f"rainfall_spread crashed with: {type(e).__name__}: {e}"
  | AssertionError: rainfall_spread crashed with: AttributeError: 'NoneType' object has no attribute 'split'
  | assert False
  | Falling test case: test_rainfall_spread_no_crash(
  |     s=None,  # or any other generated value
  | )
  +-----------------------------------
=========================== short test summary info ============================
FAILED test_weather_alert.py::test_parse_reading_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_weather_alert.py::test_wind_category_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_weather_alert.py::test_rainfall_spread_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 0.27s ===============================

(venv) E:\L6_L7_32_weather_alert\lab7>
```

**the fixed version:**

```
(venv) E:\L6_L7_32_weather_alert\lab7>python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): 35.5/80
Temperature and Humidity: (35.5, 80)

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): 45
Wind Category: advisory

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 2,5.5,10
Rainfall Spread: 8.0

(venv) E:\L6_L7_32_weather_alert\lab7>
```

```
(venv) E:\L6_L7_32_weather_alert\lab7>python weather_alert.py
==== Running parse_reading ====
Enter temperature/humidity (e.g. 35.5/80): abc
Temperature and Humidity: Invalid reading

==== Running wind_category ====
Enter a wind speed in km/h (e.g. 45): 2
Wind Category: none

==== Running rainfall_spread ====
Enter hourly rainfall in mm as CSV (e.g. 2,5.5,10): 2,5.5,10
Rainfall Spread: 8.0

(venv) E:\L6_L7_32_weather_alert\lab7>
```

